

Advanced International Macroeconomics B (EI136)

Class 6

Low interest rates

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Outline of today class

- Patterns of low level of interest rates.
 - Accounting for low sensitivity (Fiorentini et al. 2026).
 - Role of exchange rates (Bulir and Vlcek 2024).
 - A very long view (Rogoff, Rossi, and Schmelzing 2024)
- Summary of possible causes (convenience yield of government bonds, demographics, inequality).
- An overall view of the many R^* (Reis 2025).
- Drivers of asset demand and supply (Auclert et al. 2025).

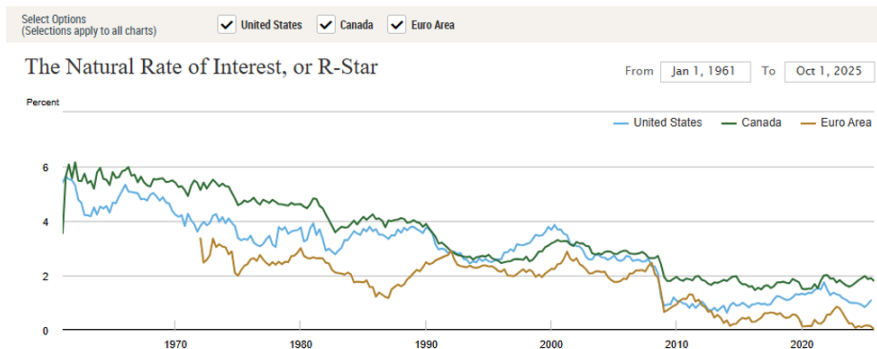
LOW INTEREST RATES : PATTERN

Estimating the natural rate

- The interest can be low because of a temporary business cycle, or because the trend interest rate has been reduced.
- Estimate the natural rate of interest r^* , that is the one at which output is at potential.
- This requires a model to discipline the data (Holston, Laubach, Williams approach - HLW).
 - Standard DSGE model with an IS (Euler curve) and a New Keynesian Phillips curve.
 - Potential output and the natural rate are not observed. They are estimated using a Kalman filter.
- Relies on output being sensitive to the interest rate (IS curve) and inflation being sensitive to the output gap (Phillips curve).

Downward trend in real rates

- Estimates of r^* from NY Fed website show a downward trend, especially since 2008.



Federal Reserve Bank of New York, measuring the natural rate of interest webpage <https://www.newyorkfed.org/research/policy/rstar>

Increasing the precision

- Method generates broad bands of confidence when the Phillips curve or the IS (Euler) curves are flat (Fiorentini et al. 2026).
 - Flat IS: output gap is not sensitive to the interest rate gap. The output gap provides limited information about non-growth drivers of r^* .
 - Flat Phillips curve: inflation is not sensitive to the output gap. It is not possible to distinguish potential output from non-growth drivers of r^* .
 - This makes the Kalman filter imprecise.
- Baseline model does not restrict the dynamics of the interest rate gap.
- Impose the additional assumption that the interest rate gap is stationary. Monetary policy does not persistently deviate from r^* .
- The point estimates for the natural rates are not altered, but the bands of confidence are reduced.

- More precise estimates when the interest rate gap is stationary.

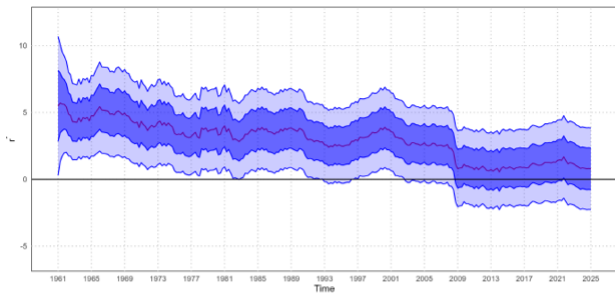


Figure 2: Natural rate of interest in the HLW23 model

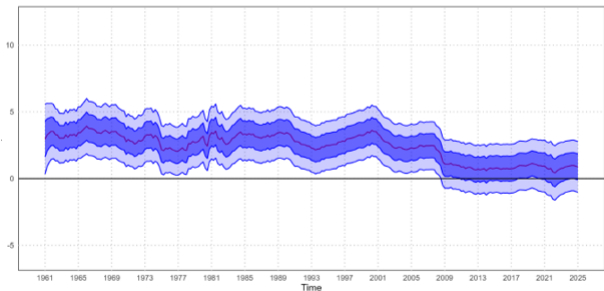


Figure 5: Natural rate of interest in the augmented model

Florentini, Gabriele, Alessandro Galesi, Rodrigo Peña, Gabriel Pérez Quirós, and Enrique Sentana (2026). "Estimating the Natural Rate of Interest under Flat IS and Phillips Curves", CEPR discussion paper 21256.

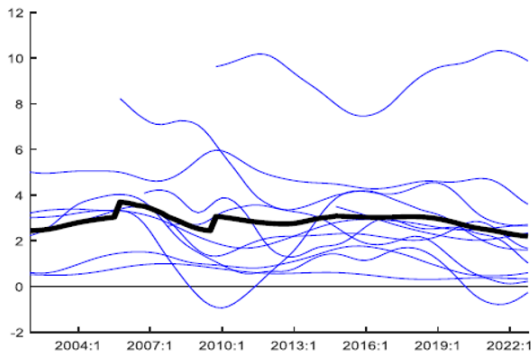
- The HLW approach has been extended to several countries, but always with a closed economy setting.
- Includes the real exchange rate.
 - In long run trends: countries catching up have a trend appreciation, which lowers r^* (trend UIP).
 - In deviations: IS curve (weak currency stimulates demand), Phillips curve (weak currency raises import costs).
 - Add UIP, and policy Taylor rule.
- Estimation for countries around the euro area, starting in 2002 (a bit of a short sample).
- r^* estimated taking account of the real exchange rate is stable, and higher than HLW.

Estimates of r^*

- Estimated values are stable (Switzerland often around 0 due to safe haven nature).

Figure 2. Time-Varying Estimates of r^* , 2002Q1–2022Q4
(In percent)

Notes: The thick black line is the unweighted mean of r^* in our 12 sample countries; the thin blue lines are individual-country estimates.



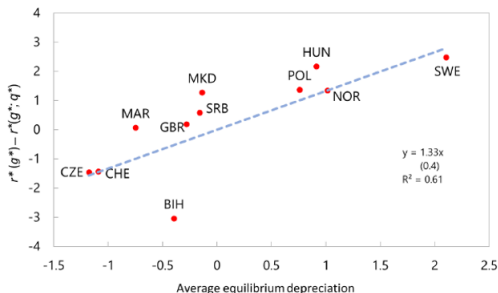
Bulir, Ales, and Jan Vilek (2024). "The Mirage of Falling R-stars", IMF working paper 2024/161

Connection with trend exchange rate

- Countries with trend real appreciation (left part of chart) have a higher r^* than inferred from HLW (negative difference, bottom part of chart).

Figure 4. Exchange Rate Appreciation and r^* Adjustment, 2013–2019
(Percentage points)

Notes: The horizontal axis measures the mean equilibrium exchange rate, with negative numbers indicating appreciation and positive numbers depreciation. The vertical axis measures the difference between the HLW (closed-economy) estimate of r^* , denoted as $r^*(g^*)$ and this paper's (open-economy) estimate of r^* , denoted as $r^*(g^*, q^*)$. The point estimate for the intercept was not significantly different from zero, hence the linear regression shown on the chart is estimated without the intercept; the standard error of the point estimate is in parentheses. The sample excludes Türkiye, although its inclusion does not change the slope and increases the coefficient of determination only marginally.



Bulir, Ales, and Jan Vlcek (2024). "The Mirage of Falling R-stars", IMF working paper 2024/161

Taking a long view

- Historical data on interest rates on government bonds since 1870 (Del Negro et al. 2019).
 - Equilibrium real interest rate has been steady until the 1980's (some decrease in the 1950's and 1960's).
 - Clear downward trends since the 1980's. This pre-dates the global financial crisis.
- A very long view from the 14th century on (Rogoff, Rossi, and Schmelzing 2024).
 - Long maturity bonds and inflation (urban prices).
- Steady downward trend in real interest rates (1.7 p.p. per century).
 - Stationarity around that trend.
- Little evidence for breaks (except black death and large defaults of 1557 in France, Spain, Netherlands).
- Real interest rates not correlated with population or GDP growth.
- Trend: rising liquidity, rising demand for safety, decreasing default risk.

Trend in real rates

- Downward trend since 1300, recent plateau.

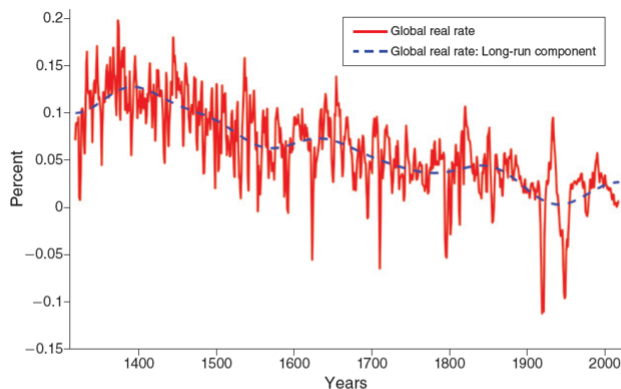


FIGURE 1. HEADLINE GLOBAL REAL RATES AND ITS LONG-RUN COMPONENT, 1318–2022

Notes: Data based on Schmelzing's (2023) GDP-weighted global real rate data. Long-maturity ex ante basis, deflated by using seven-year progressively lagged inflation, excluding current year t . The dashed line reports the long-run filtered component, isolating fluctuations above 100 years, based on the Müller and Watson (2018) estimator.

Rogoff, Kenneth, Barbara Rossi, and Paul Schmelzing (2024). "Long-Run Trends in Long-Maturity Real Rates, 1311–2022", *American Economic Review* 114(8), pp. 2271–2307.

- No connection with GDP and population growth rates (going opposite ways).

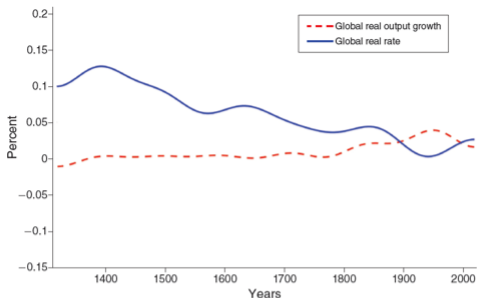


FIGURE 2. A COMPARISON OF MÜLLER-WATSON-FILTERED REAL AGGREGATE OUTPUT GROWTH RATES AND REAL INTEREST RATES, 1318–2021

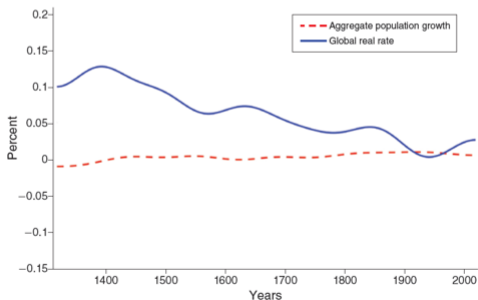


FIGURE 3. A COMPARISON OF MÜLLER-WATSON-FILTERED AGGREGATE POPULATION GROWTH RATES AND REAL INTEREST RATES, 1318–2021

Rogoff, Kenneth, Barbara Rossi, and Paul Schmelzing (2024). “Long-Run Trends in Long-Maturity Real Rates, 1311–2022”, *American Economic Review* 114(8), pp. 2271–2307.

EXPLANATIONS

Convenience yields

- Studies focus on the interest rate on Treasury bonds. They are “special”, as benchmark assets (safety, liquidity).
- Trend Treasury yield fell by 1.0 to 1.4 percent from 1998 to 2016, depending on method.
- Special nature (convenience yield) estimated from spread Baa corporate - Treasury (safety and liquidity components).
 - Accounts for 80-90 bps, primarily from liquidity.
- Rising convenience can reflect foreign demand, higher need of collateral for securitization, higher regulatory requirement on financial intermediaries

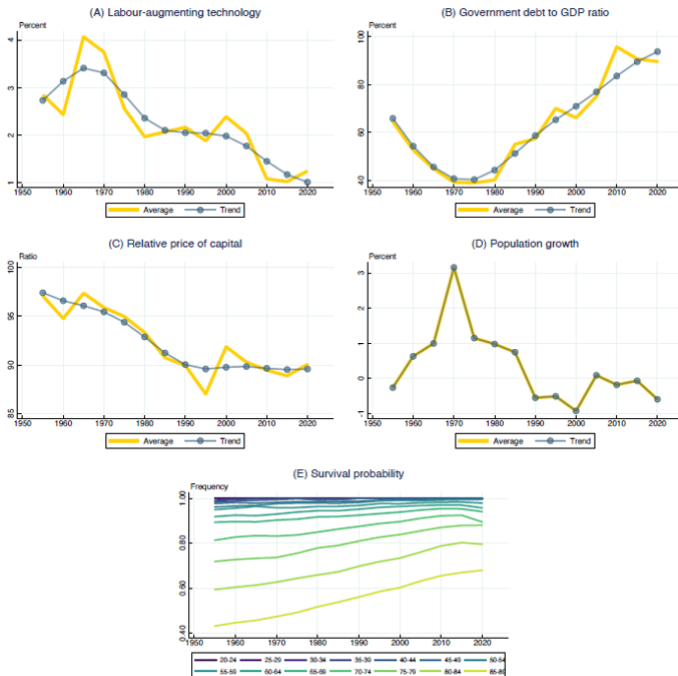
- Substantial changes in structure of the population since the 1960's (Lisack et al. 2021).
 - Temporary increase in births (baby boom) and rising life expectancy.
 - Aging is broad-based and more pronounced in Europe and emerging markets..
- Baby-boomers boosted labor supply when they were in middle age (lower K/L ratio), and are now going into retirement.
- OLG model with changing demographics. Impact on interest rate and wealth due to behavior and composition effects.
 - Behavior: long life expectancy $\rightarrow$ more time in retirement $\rightarrow$ accumulate assets.
 - Composition: households aged 50 and above are wealthier.
- Aging leads to high savings, high K/L , hence low real interest rate.

Contrasting demographics and other factors

- Assess demographics (population growth, and higher life expectancy), changing trend productivity, higher public debt, changing price of capital (Cesa-Bianchi et al. 2022).
 - Global trends since 1955, and country-specific elements.
- OLG model to assess the impact on interest rates.
 - Expectations adjust gradually to changing trends.
- Global r^* increases from 1.25% in 1955 to 2.75% by 1975, followed by decrease to -0.25 in 2015 (similar to estimates from VAR, with earlier peak in model).
- Decrease driven by lower productivity growth, and higher life expectancy (similar result in Christensen and Pedersen 2022).

- Slowing productivity, higher life expectancy.

Figure 2: Global trends in the five drivers

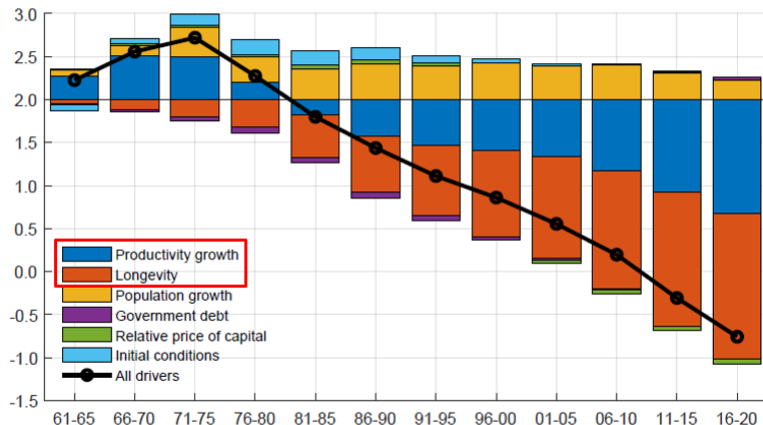


Cesa-Bianchi, Ambrogio, Richard Harrison, and Rana Sajedi (2023). "Global R^* ", CEPR Discussion Paper 18518.

Decomposition of interest rate

- Productivity growth and life expectancy are the key factors.

Figure 6: Decomposition of the drivers of Global R^*



Cesa-Bianchi, Ambrogio, Richard Harrison, and Rana Saiedi (2023). "Global R^* ", CEPR Discussion Paper 18518.

- Inequality, with large shift in the US income distribution (Mian et al 2021).
 - Rising share of income to top groups, with higher savings rates.
 - Saving rate is more heterogeneous across incomes than across ages.
- Monetary policy in core countries (Borio et al. 2022).
- Panel of real interest rates drivers (19 countries since 1870)
 - Demographics and productivity are relevant, but only in recent decades.
 - Global factors matter more than domestic ones.
 - Interest rates in core countries (UK pre 1914, then US). Taylor rule residuals.
 - Robust driver of real rates in other countries.

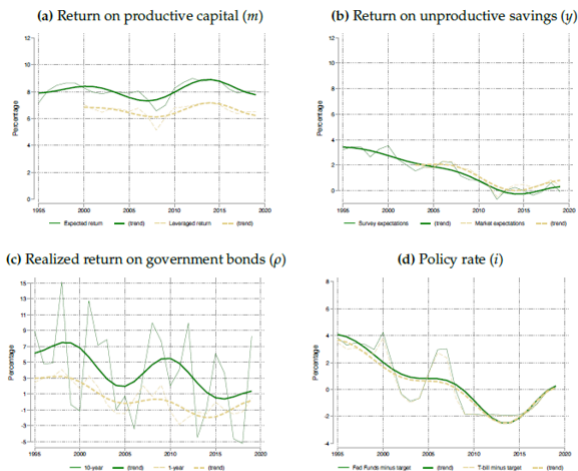
FOUR CONCEPTS OF R^*

The various interest rates

- Four real interest rates are relevant (Reis 2025).
 - Marginal return on private capital, m .
 - Yield on long term sovereign bond, y . Reflects expected inflation.
 - Realized return on sovereign bond on a shorter period, ρ . Reflects inflation surprises and varying bond price (if sold before maturity).
 - Policy interest rate i .
- Heterogeneous evolution of the four rates.
 - Steady m (profits / capital, can be augmented by spreads).
 - Declining y (10 year yield adjusted by inflation forecast, TIPS yield), ρ (realized return on 1 year holding), and i (Fed funds).

Evolution of the four real rates.

Figure 1: The four R-stars and their trends between 1995 and 2019

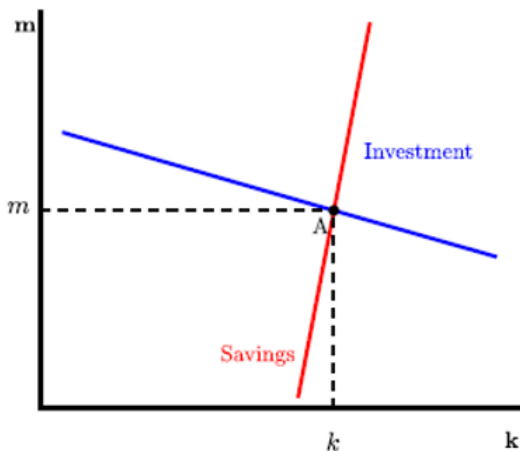


Notes: Panel (a): Expected return is the US nominal return on private capital minus the average of expected 10-year-ahead inflation in the SPF. Leveraged return is a weighted average cost of capital using Baa bonds and effective tax rates. Panel (b): "Survey expectation subtracts from the 10-year yield on US government bonds the average expected 10-year-ahead inflation in the SPF, while market expectations is the yield on a 10-year Treasury inflation-protected security. Panel (c): Holding period real return on 10-year and 1-year US government bonds. Panel (d): Federal Funds rate or 3-month government bond rate minus 2%. Trends calculated using Mueller-Watson filter with a 10-year window.

Reis, Ricardo (2025) "The Four R-stars: From Interest Rates to Inflation and Back", mimeo.

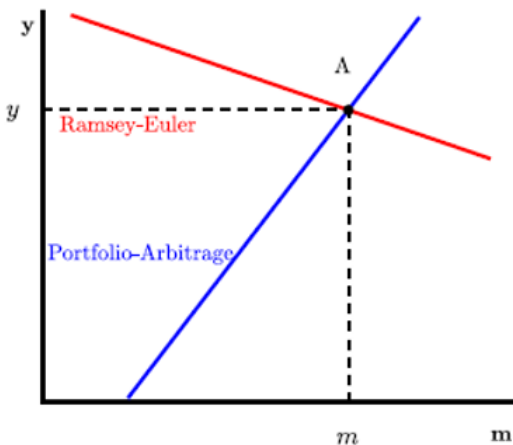
- Link capital (investment) k and the marginal return m .
- Downwards sloping **investment line**: higher k leads to a lower m . Shifted by productivity, public investment.
- Upwards sloping **savings line**: higher return draws in savings.

(a) Productive investment and savings



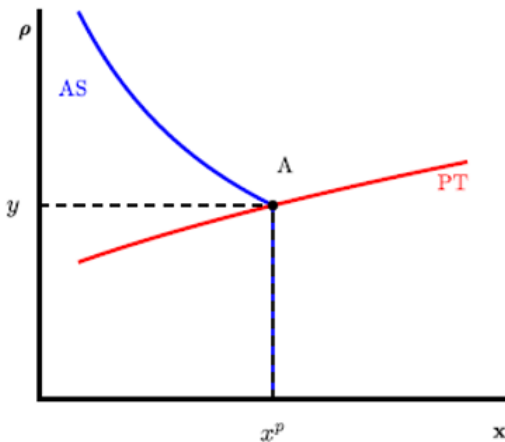
- Link yield on bonds y and the marginal return m .
- Upwards sloping **no-arbitrage line**: return equalized, with a convenience component on bonds. Shifted by preference for safety.
- Downwards sloping **Ramsey line**: Euler calls for a given overall return. Higher return on one asset offset by lower return on the other.

(b) Unproductive or productive savings



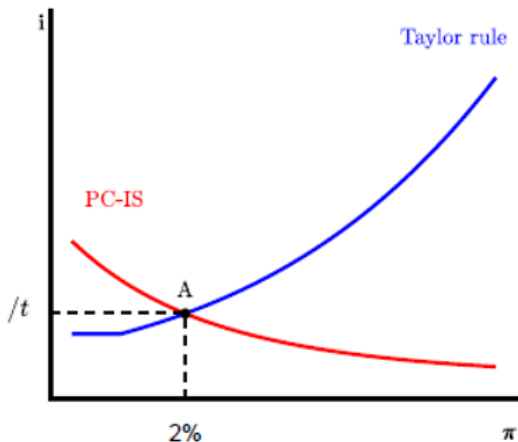
- Link realized returns ρ and output gap x .
- Downwards sloping **aggregate supply**: unexpected inflation reduces ρ below y and puts output at potential (vertical segment), low inflation surprises raises ρ but leads to recession (low x).
- Upwards sloping **policy line**: trade off with adverse supply shock, output below potential and high inflation (low realized return).

(c) Realized returns and policy choices



- Link policy rate i and inflation π .
- Upwards sloping **Taylor rule**: raise interest rate in the face of inflation.
- Downwards sloping **IS line**: interest rate below the neutral short term (long term rate y scaled by term premium t) puts inflation above target.

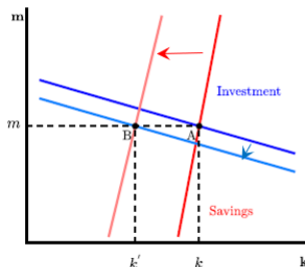
(d) Monetary policy



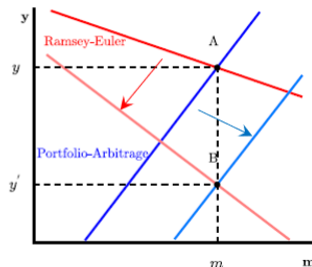
The pre-Covid trends (1)

- Capital and marginal return (panel a).
 - Reduction in investment demand (lower investment prices, lower long-term growth, lower public investment).
 - Reduction in saving supply.
- Bond yields and capital return (panel b).
 - Raising savings supply (aging, foreign glut) leads to accept lower return (Ramsey down).
 - Preference shift towards bonds, higher convenience yield.

(a) Productive investment and savings



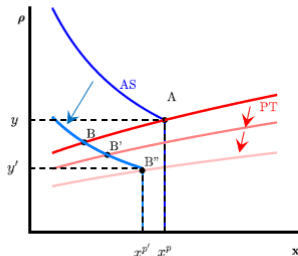
(b) Unproductive or productive savings



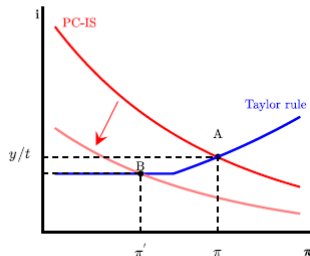
The pre-Covid trends (2)

- Output gap and realized return (panel c).
 - Lower capital shifts AS left, and lower y lowers the “corner”.
 - Policy initially unchanged, leading to low inflation (ρ is above the new y) and recession. Gradual adjustment of policy (fiscal and QE).
- Inflation and policy interest rate (panel d).
 - Lower yield y moves down the IS curve.
 - Inflation falls more when the Taylor rule is constrained by the lower bound.

(c) Realized returns and policy choices



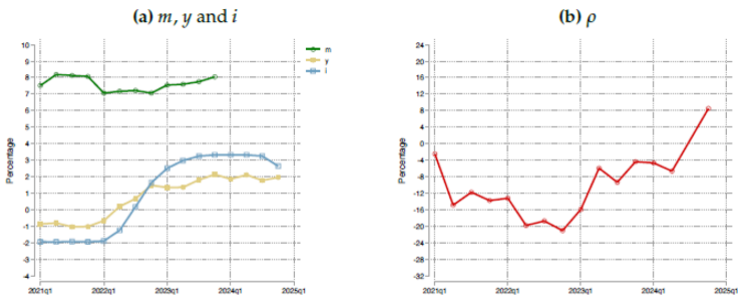
(d) Monetary policy



Reis, Ricardo (2025) “The Four R-stars: From Interest Rates to Inflation and Back”, mimeo.

- Narrowing of $m - y$ gap, as yields and policy rates increased. Inflation lowered realized return ρ .
- Can be explained by the 4 panel framework.
- Uncertain scenarios for the future. Specific source of the narrowing $m - y$ matters.

Figure 5: The four returns since 2021



Notes: see the notes in figure 1. For m this is the unleveraged return, for y it is the market yield, for ρ it is the annual holding return on a 10-year bond, and for i it is the Federal funds rate.

ASSETS DEMAND AND SUPPLY

- Rising wealth to GDP (U.S.) since the 1980's, with downward trend in interest rate since 1950 (Auclert et al. 2025).
- Contrast the role of asset demand and supply, and their drivers.
- Asset demand: households and rest of the world.
 - Demographics: older people hold more assets, longer lives increase need to save.
 - Public retirement: more generous social security reduces need to save.
 - Inequality: people with higher income have more assets.
- Asset supply: government, capital stocks, and NPV of profits from markups.
 - Rising markups increase equity prices and asset value.
- Initial contraction of supply (lower profits), and subsequent increase in demand.

Empirical pattern

- Trends in asset / GDP and real interest rate.

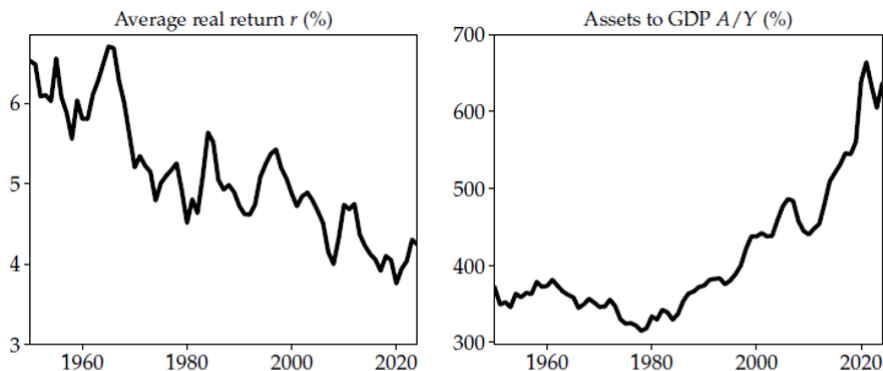


Figure 2: Falling returns and rising wealth, 1950–2024

Auclert, Adrien, Hannes Malmberg, Matthew Rognlie, and Ludwig Straub (2025). "The Race Between Asset Supply and Asset Demand", paper presented at the IMF 2025 Jackson Hole symposium.

- Supply of assets initially contracted (low profits due to low productivity), and then increased through higher profits.
- Demand reflects households.
 - Limited changes until 1970. Lower supply is the main driver until 1980, followed by rising demand.
- Model with varying drivers.
 - Government debt, productivity and markup changes on the supply side.
 - Demographics (OLG) and inequality on the demand side.
- Calibration of drivers shows:
 - Weak productivity reduces supply initially, higher markup increased it in the 21st century.
 - Demographics increase demand (offset in part by social security), as does inequality since 2000.

- Demand from households, supply from profits.
- Initially, large role of supply, followed by demand.

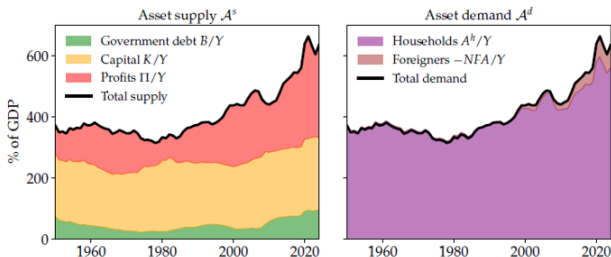


Figure 3: Realized components of asset supply and demand, 1950–2024

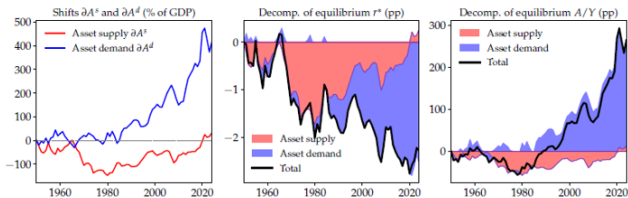
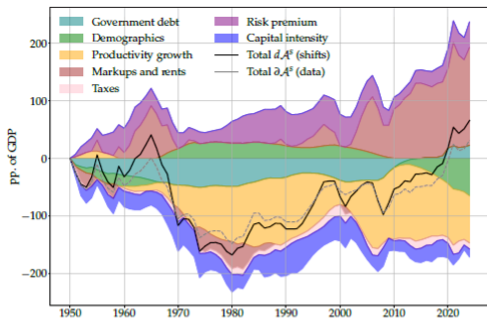


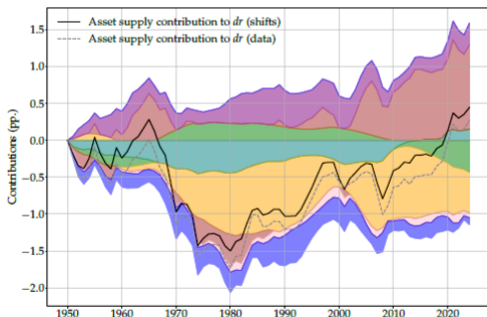
Figure 4: Decomposing asset market equilibrium into demand and supply shifts

Auclert, Adrien, Hannes Malmberg, Matthew Rognlie, and Ludwig Straub (2025). "The Race Between Asset Supply and Asset Demand", paper presented at the IMF 2025 Jackson Hole symposium.

- Weak 1970's productivity reduced supply.
- Higher markups increased it since 2000.



(a) Asset supply



(b) Interest rate

Figure 9: Asset supply shifts and effects on interest rate

Auclert, Adrien, Hannes Malmberg, Matthew Rognlie, and Ludwig Straub (2025). "The Race Between Asset Supply and Asset Demand", paper presented at the IMF 2025 Jackson Hole symposium.

- Demographics raised demand, partially offset by more generous social security.
- Productivity growth raises income and asset demand.
- Higher inequality since 2000.

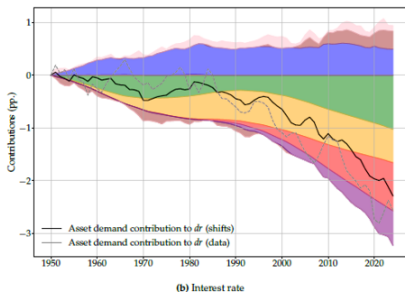
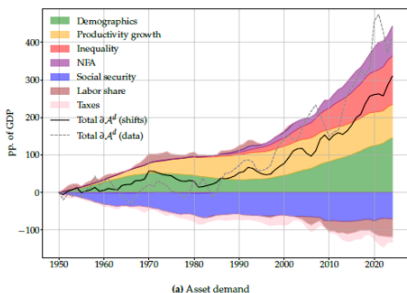


Figure 10: Asset demand shifts and effects on interest rate

Auercl, Adrien, Hannes Malmberg, Matthew Rognlie, and Ludwig Straub (2025). "The Race Between Asset Supply and Asset Demand", paper presented at the IMF 2025 Jackson Hole symposium.

- Continuing ageing (and inequality) will keep increasing demand.
 - At given supply, the interest rate should decrease by 1.2 percent.
- Public deficits, including Medicaid and Medicare, lead to higher debt and supply.
 - At given demand, the interest rate should increase by 1.7 percent.
- Asset supply stronger than demand, so equilibrium requires fiscal consolidation.